

Artavazd Maranjyan

✉ arto.maranjyan@gmail.com

🌐 artomaranjyan.github.io

🔗 [Google Scholar](#) [LinkedIn](#)

Current Position

2026–present **Postdoctoral Researcher**, working with Volkan Cevher and Martin Jaggi, EPFL, Lausanne, Switzerland

Education

- 2023–2025 **Ph.D. in Computer Science**, KAUST, Thuwal, Saudi Arabia; **Advisor:** Peter Richtárik
■ **Thesis:** [First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data](#)
- 2021–2023 **M.Sc. in Applied Statistics and Data Science**, Yerevan State University, Yerevan, Armenia
■ **Thesis:** [On local training methods](#); **Co-supervisors:** Peter Richtárik, Mher Safaryan
- 2017–2021 **B.Sc. in Informatics and Applied Mathematics**, Yerevan State University, Yerevan, Armenia
■ **Thesis:** [On the Convergence of Series in Classical Systems](#); **Supervisor:** Martin Grigoryan

Academic Experience

- 03/2025 **Research Visit to Yi-Shuai Niu**, BIMS, Beijing, China
○ Gave talks at three universities (PKU, BUAA, BIMS)
○ Collaborated with Professor Yi-Shuai Niu on a project on Server-Assisted Federated Learning
- 04/23–08/23 **Researcher in the Group of Martin Grigoryan**, Yerevan State University, Yerevan, Armenia
○ Studied the existence and properties of universal functions with respect to the Vilenkin and Haar systems across various functional spaces
- 06/22–01/23 **Internship in the Group of Peter Richtárik**, KAUST, Thuwal, Saudi Arabia
○ Worked on the "GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity" paper
- 01/22–08/23 **Machine Learning Researcher**, YerevaNN, Yerevan, Armenia
○ Worked on the intersection of Federated Learning and Optimization

Industry Experience

- 07/21–06/22 **Co-Founder**, [OnePick](#), Yerevan, Armenia
○ Built an MVP platform to help content creators and marketers generate or select photos that best matched their text descriptions
○ Participated in the [InVent 2.0 \(FAST\)](#) venture building program, advancing through team formation, idea generation, product development, and pitching
○ Secured initial funding and continued development beyond the program
- 07/21–09/21 **Backend Developer**, EXALT Technologies Ltd, Yerevan, Armenia
○ Worked for Nutanix
- 06/21–07/21 **Machine Learning Research Engineer**, FAST, Yerevan, Armenia
○ Worked on Fraud detection
○ Made data-driven forecasts using machine learning algorithms and statistical models
- 09/19–01/21 **Software Engineer in Test**, Picsart, Yerevan, Armenia
○ Worked with automation team to design and develop automated solutions across several mobile/web applications
○ Worked directly with software developers, test engineers, product owners, business analysts to find and resolve issues
○ Worked closely with DevOps to suggest improvements in processes and in Jenkins Continuous Integration cycle

Awards

- 06/2026 **EPFL AI Center and Swiss AI Initiative Postdoctoral Fellowships**, EPFL
Competitive fellowship supporting independent postdoctoral research, with additional travel funding

- 05/2025 **CEMSE Dean's List Award, KAUST**
Awarded for excellent academic and research performance (\$2,500 prize)
- 09/2023 **Dean's Award, KAUST**
Awarded to a few top students accepted to KAUST (\$6,000 annually for 3 years)
- 05/2021 **Outstanding Final Project Award, Yerevan State University**
Recognized for the Bachelor's thesis (awarded to 6 students among 250+ students)

Publications

15. LOSCAR-SGD: Local SGD with Communication-Computation Overlap and Delay-Corrected Sparse Model Averaging. ■ Y. Maziane, A. Mahran, A. Maranjyan, P. Richtárik. *arXiv:2605.20866*, 2026.
14. Ringmaster LMO: Asynchronous Linear Minimization Oracle Momentum Method. ■ A. Sadiev, A. Maranjyan, I. Ilin, P. Richtárik. *arXiv:2605.18174*, 2026.
13. Rescaled Asynchronous SGD: Optimal Distributed Optimization under Data and System Heterogeneity. ■ A. Mahran, A. Maranjyan, P. Richtárik. *arXiv:2605.13434*, 2026.
12. Rennala MVR: Improved Time Complexity for Parallel Stochastic Optimization via Momentum-Based Variance Reduction. ■ Z. Tovmasyan, A. Maranjyan, P. Richtárik. *arXiv:2605.08871*, 2026.
11. Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity. ■ A. Maranjyan, P. Richtárik. *ICLR* 2026.
10. BiCoLoR: Communication-Efficient Optimization with Bidirectional Compression and Local Training. ■ L. Condat, A. Maranjyan, P. Richtárik. *arXiv:2601.12400*, 2026.
9. ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning. ■ A. Maranjyan, E. M. Saad, P. Richtárik, F. Orabona. *ICML* 2025.
8. Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity. ■ A. Maranjyan, A. Tyurin, P. Richtárik. *ICML* 2025.
7. MindPlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times. ■ A. Maranjyan, O. S. Omar, P. Richtárik. *UAI* 2025; *OPT* 2024. **Oral (top 5%)**.
6. GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity. ■ A. Maranjyan, M. Safaryan, P. Richtárik. *TMLR 2025: Transactions on Machine Learning Research*.
5. LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression. ■ L. Condat, A. Maranjyan, P. Richtárik. *ICLR* 2025. **Spotlight (top 5.1%)**.
4. Differentially Private Random Block Coordinate Descent. ■ A. Maranjyan, A. Sadiev, P. Richtárik. *OPT* 2024.
3. Menshov-type theorem for divergence sets of sequences of localized operators. ■ M. Grigoryan, A. Kamont, A. Maranjyan. *Journal of Contemporary Mathematical Analysis*, vol. 58, no. 2, pp. 81–92, 2023.
2. On the divergence of Fourier series in the general Haar system. ■ M. Grigoryan, A. Maranjyan. *Armenian Journal of Mathematics*, vol. 13, pp. 1–10, 2021.
1. On the unconditional convergence of Faber-Schauder series in L^1 . ■ T. Grigoryan, A. Maranjyan. *Proceedings of the YSU A: Physical and Mathematical Sciences*, vol. 55, no. 1 (254), pp. 12–19, 2021.

Reviewing

- 2024–present TMLR (Transactions on Machine Learning Research)
- 2026 NeurIPS (Conference on Neural Information Processing Systems)
- 2026 ICML (International Conference on Machine Learning)
- 2025 ICLR (International Conference on Learning Representations)
- 2024 SIMODS (SIAM Journal on Mathematics of Data Science)
- 2024 JMLR (Journal of Machine Learning Research)

Talks and Poster Presentations

- 26/04/2026 **Protocol Learning: Decentralized Collaborative Learning at Scale (ICLR 2026 Workshop)**, Lagune Barra Hotel, Rio de Janeiro, Brazil
Delivered a talk titled **Controlling Delay in Asynchronous SGD: Optimality for Any Data Regime** [video] [slides]

- 23/04/2026 **(ICLR 2026) The Fourteenth International Conference on Learning Representations**, Riocentro Convention and Event Center, Rio de Janeiro, Brazil
Presented a poster on **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [poster]
- 09/02/2026 **KAUST Rising Stars in AI Symposium 2026**, KAUST, Thuwal, Saudi Arabia
Presented posters on
- **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [poster]
 - **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [poster]
 - **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [poster]
- 13/11/2025 **AMCS/STAT Graduate Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk titled **First Provably Optimal Asynchronous SGD for Homogeneous and Heterogeneous Data** [slides]
- 04/11/2025 **The KAUST 2025 Workshop on Statistics**, KAUST, Thuwal, Saudi Arabia
Presented posters on
- **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [poster]
 - **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [poster]
- 28/10/2025 **Mathematics and Applications Colloquium**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on **Ringleader ASGD: The First Asynchronous SGD with Optimal Time Complexity under Data Heterogeneity** [slides]
- 29/07/2025 **35^o Colóquio Brasileiro de Matemática**, IMPA, Rio de Janeiro, Brazil
Delivered a talk and a poster on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides] [poster]
- 23/07/2025 **41st Conference on Uncertainty in Artificial Intelligence**, Rio Othon Palace, Rio de Janeiro, Brazil
Presented a poster on **MindFlayer SGD: Efficient Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [poster]
- 16/07/2025 **Forty-Second International Conference on Machine Learning**, Vancouver Convention Center, Canada
Presented posters on
- **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [poster]
 - **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [poster]
- 18/05/2025 **Stochastic Numerics and Statistical Learning**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [poster]
- 26/04/2025 **The Thirteenth International Conference on Learning Representations**, Singapore EXPO, Singapore
Presented a poster on **LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression** (Spotlight presentation (top 5.1% of the submitted papers))
- 16/04/2025 **FLOW Talk #126**, Federated Learning One World Seminar (FLOW), Online
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [video] [slides]
- 08/04/2025 **KAUST Rising Stars in AI Symposium 2025**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **ATA: Adaptive Task Allocation for Efficient Resource Management in Distributed Machine Learning** [poster]
- 28/03/2025 **Machine Learning Reading Group**, YSU Krisp-AI Lab, Yerevan, Armenia
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]
- 26/03/2025 **Flower AI Summit 2025**, King's House, London, England
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]
- 17/03/2025 **Academic Report of the School of Mathematical Sciences**, Beihang University (BUAA), Beijing, China
Invited by **Jiixin Xie** to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]
- 13/03/2025 **Optimization Seminar**, BIMSA, Beijing, China
Invited by **Yi-Shuai Niu** to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]

- 12/03/2025 **Seminar**, Peking University, Beijing, China
Invited by **Kun Yuan** to give a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]
- 27/02/2025 **AMCS/STAT Graduate Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on **Ringmaster ASGD: The First Asynchronous SGD with Optimal Time Complexity** [slides]
- 15/12/2024 **Workshop on Optimization for Machine Learning (NeurIPS 2024)**, Vancouver Convention Center, Vancouver, Canada
Presented
 - o **MindPlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** (Oral presentation, top 5% of 107 submissions) [video]
 - o **Differentially Private Random Block Coordinate Descent**
 - o **LoCoDL: Communication-Efficient Distributed Learning with Local Training and Compression**
- 21/11/2024 **MLR Weekly Seminar**, Machine Learning Research at Apple, Online
Invited by **Samy Bengio** to give a talk on **MindPlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [slides]
- 18/10/2024 **International Conference on Algebra, Logic, and their Applications**, Yerevan State University, Online
Delivered a talk on **MindPlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
- 10/10/2024 **CEMSE E-Poster Competition**, KAUST, Thuwal, Saudi Arabia
Awarded 3rd place for presenting a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity**
- 06/07/2024 **Analysis, PDEs and Applications**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **MindPlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times** [abstract]
- 27/05/2024 **Stochastic Numerics and Statistical Learning**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 05/05/2024 **CS 331: Stochastic Gradient Descent Methods**, KAUST, Thuwal, Saudi Arabia
Delivered a guest lecture on **MindPlayer: Efficient Asynchronous Parallel SGD in the Presence of Heterogeneous and Random Worker Compute Times**
- 13/03/2024 **The Machine Learning Summer School in Okinawa 2024**, OIST, Okinawa, Japan
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 21/02/2024 **KAUST Rising Stars in AI Symposium 2024**, KAUST, Thuwal, Saudi Arabia
Presented a poster on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [poster]
- 16/11/2023 **Group Seminar**, KAUST, Thuwal, Saudi Arabia
Delivered a talk on **Differentially Private Coordinate Descent for Composite Empirical Risk Minimization**
- 06/10/2023 **Algorithms & Computationally Intensive Inference seminars**, University of Warwick, Coventry, England
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [slides]
- 05/07/2023 **Mathematics in Armenia: Advances and Perspectives**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [abstract]
- 10/03/2023 **Machine Learning Reading Group**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [video (Yerevan, Armenia)]
- 07/12/2022 **FLOW Talk #88**, Federated Learning One World Seminar (FLOW), Online
Delivered a talk on **GradSkip: Communication-Accelerated Local Gradient Methods with Better Computational Complexity** [video]
- 10/04/2022 **Machine Learning Reading Group**, Yerevan State University, Yerevan, Armenia
Delivered a talk on **ProxSkip: Yes! Local Gradient Steps Provably Lead to Communication Acceleration! Finally!**